Structurally Forced Information Divergence via Colexification Asymmetries

Interim findings from CLICS4 analysis

1. Motivation

When two languages describe the same situation, some differences in their output are editorial (what the author chose to include) and some are structural (what the language forces you to commit to or leave open). X-PARADE (Rodriguez et al., 2024) annotates the former. This project targets the latter.

The mechanism: if language A has one word covering concepts X and Y, a speaker of A can use that word without specifying which they mean. A speaker of language B, which uses separate words for X and Y, must choose. That choice is forced by lexical structure, not editorial preference.

CLICS4 provides a catalogue of where these conceptual boundary splits occur across approximately 3,400 language varieties and 247 language families. We use it to identify, for any language pair, exactly where forced divergences should arise.

2. Data

CLICS4 contains 1,730 concepts, 3,447 language varieties, and 1.45 million word form entries. Colexification is detected by matching on phonological form (IPA segments). Each colexification edge records which languages and families attest it. Of 51,562 concept pairs colexified in at least one language, 321 are colexified across 20 or more language families. The interesting zone for forced divergence is pairs colexified in many families but not all.

3. Most promising concept pairs

Concept pair	Families colexifying	Colexify (major langs)	Distinguish (major langs)
LEG / FOOT	100	Russian, Japanese, Bengali, Turkish, Swahili	English, Spanish, French, German, Arabic, Korean
DO / MAKE	76	Spanish, French, Portuguese, Italian, Korean, Mandarin, Turkish, Vietnamese	Japanese, Russian, Polish
WOMAN / WIFE	59	Spanish, Portuguese, German, Dutch, Arabic	English, Russian, French, Japanese, Korean
FLESH / MEAT	105	Spanish, Japanese, Mandarin, German, Dutch, Vietnamese	English, Arabic
HEAR / LISTEN	53	Japanese, Korean, Bengali, Vietnamese	English, Spanish, French, German, Russian
MOON / MONTH	60	Japanese, Korean, Turkish, Swahili, Thai	English, Spanish, French, Russian, German
BLUE / GREEN	70	(various, mostly non-major NLP langs)	English, Spanish, French, Japanese, Korean, Mandarin

4. LEG / FOOT: the strongest case

Russian "noga" covers both leg and foot. As Cornell's Russian lexical database states: "Russian differs from English in that the word noga may mean either leg or foot." This produces forced divergence in translation constantly.

Naturally occurring examples from parallel text:

> **U menya bolit noga** -> My leg hurts / My foot hurts (both valid)
>
> **Sobaka ukusila yeyo za nogu** -> A dog bit her leg (could be foot)
>
> **Ya slomal pravuyu nogu** -> I broke my right leg (could be right foot)
>
> **Novyye botinki natyorli nogi** -> New boots rubbed one's feet sore (context disambiguates toward feet)
>
> **Ona sidela s nogami v uglu divana** -> She was sitting with her legs tucked (context disambiguates toward legs)

In the first three examples, the Russian is genuinely underspecified; the English translator must commit. In the last two, context disambiguates. The forced divergence cases are those where context does not resolve the ambiguity.

5. Connection to X-PARADE

In the X-PARADE taxonomy, the structural commitment forced by the target language is new information: content in the target paragraph not present in the source, introduced because the target language requires a distinction the source language does not make. But unlike the editorial new information in X-PARADE (different facts chosen by different Wikipedia editors), this new information is linguistically forced and systematically predictable from CLICS data.

6. Next steps

Phase 2: Search parallel corpora (OpenSubtitles, Tatoeba, literary translations) for sentences where the colexifying word is used in context that does not disambiguate. Focus on LEG/FOOT (Russian-English, Japanese-English), WOMAN/WIFE (Spanish-English), and HEAR/LISTEN (Japanese-English).

Phase 3: Test LLM consistency on these examples. When an LLM translates the ambiguous source sentence, which target word does it choose? Is the choice consistent across runs and models? When asked about the content in both languages, does the LLM give compatible answers?

Phase 4: Quantify how much of the information divergence in cross-lingual text is attributable to colexification asymmetries versus editorial choice. CLICS provides the prior expectation; corpus data provides the observation.